COSC264
Introduction to Computer Networks and the Internet

# The Network Layer in the Internet

Dr Barry Wu

WRC
University of Canterbury
Barry.wu@canterbury.ac.nz

# Learning Objectives

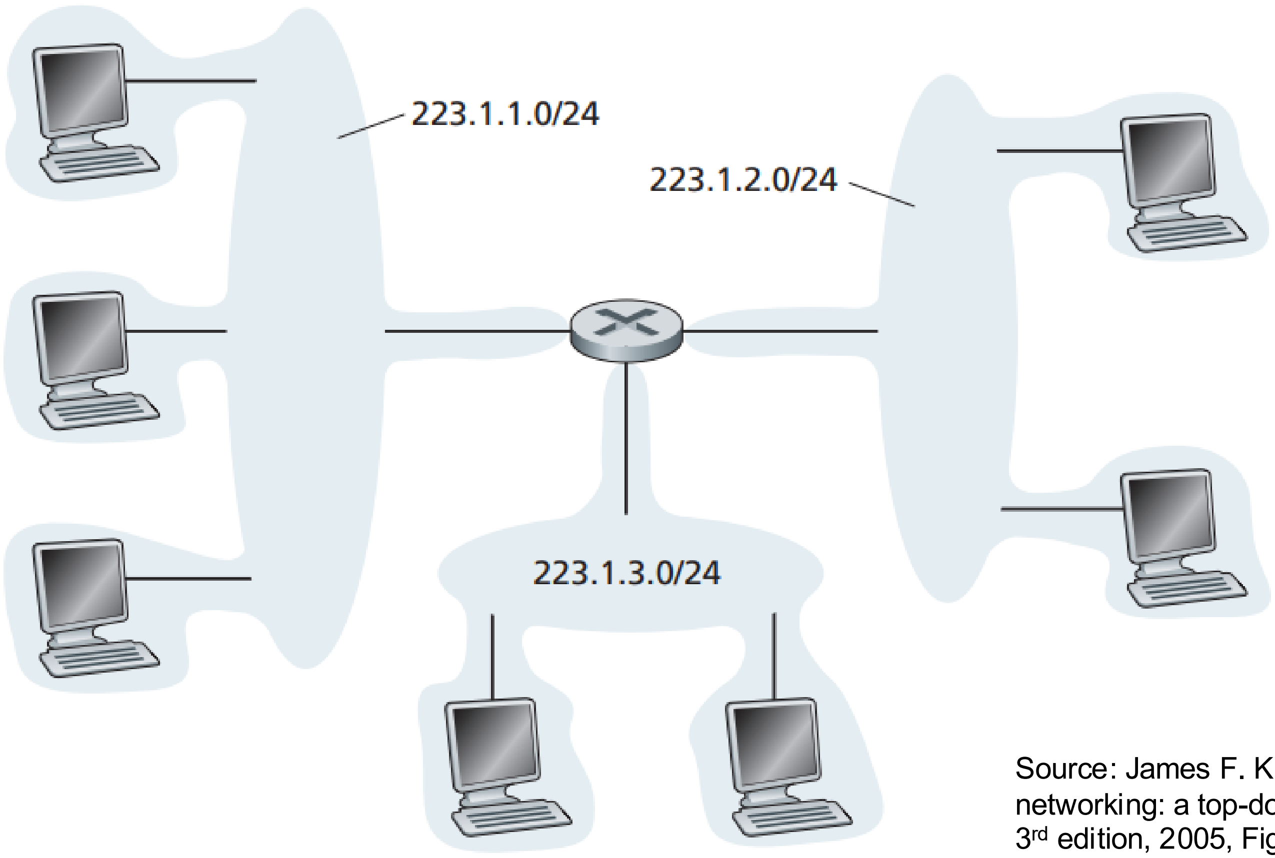
- At the end of the lectures, you will be able to:
  - Describe and compare the three routing protocols, including BGP, RIP, and OSPF
  - Understand the network address translation
  - Understand the IP version 6
  - Understand address aggregation

# Outline

- Routing in the Internet
  - BGP
  - RIP
  - OSPF
- NAT
- IPv6
- Address aggregation

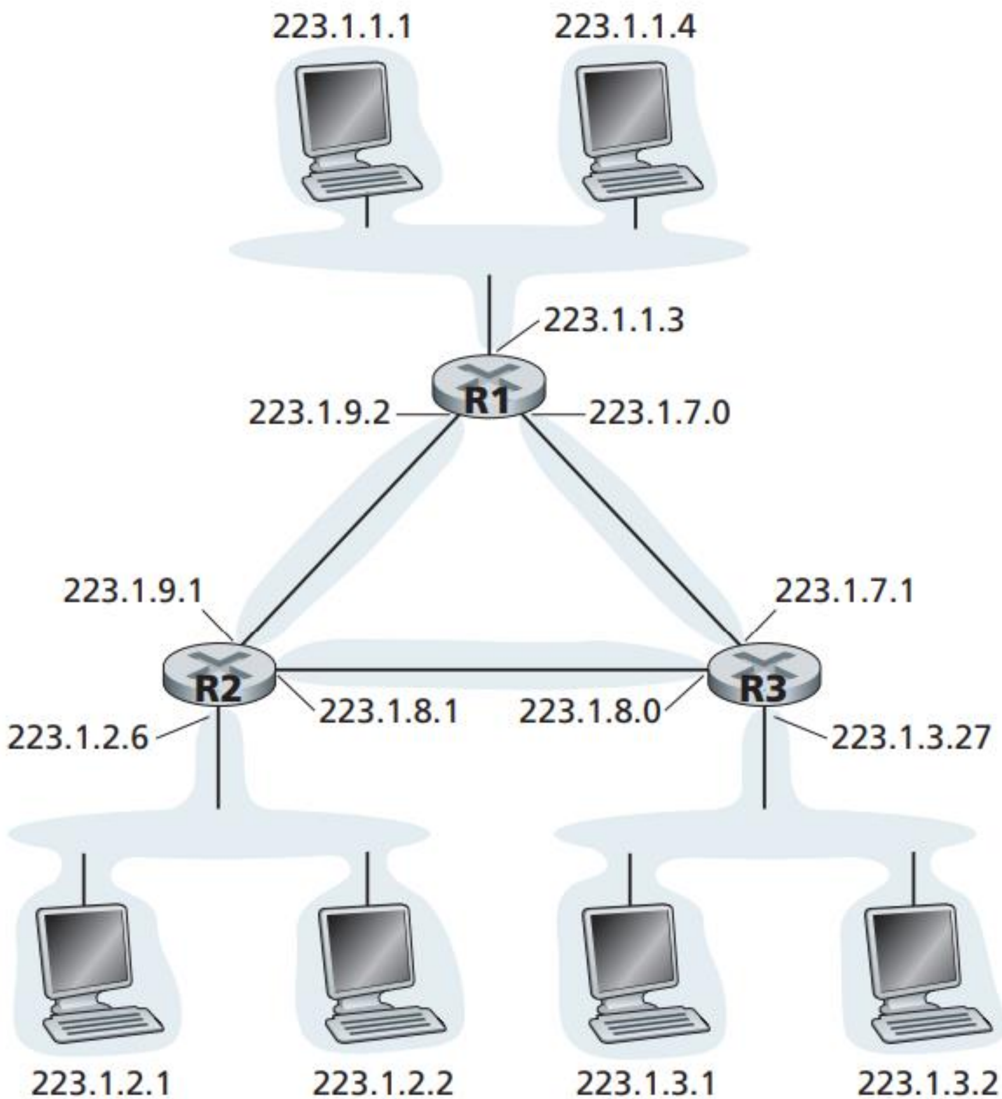
# Subnets

- To determine the subnets, detach each interface from its host or router, creating islands of isolated networks, with interfaces terminating the end points of the isolated networks. Each of these isolated networks is called a subnet.




Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 4.16.

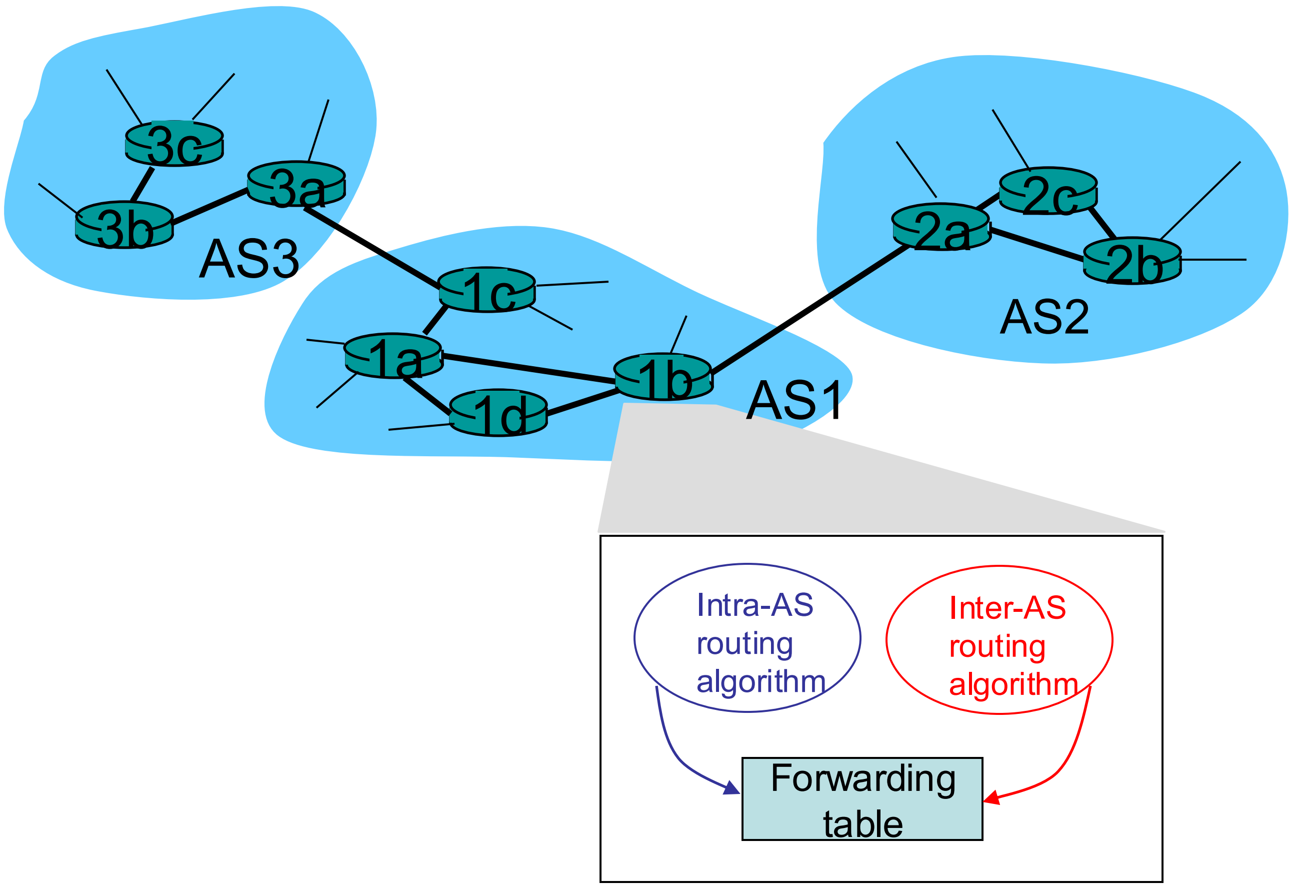
# Subnets (cont.)




6 subnets!

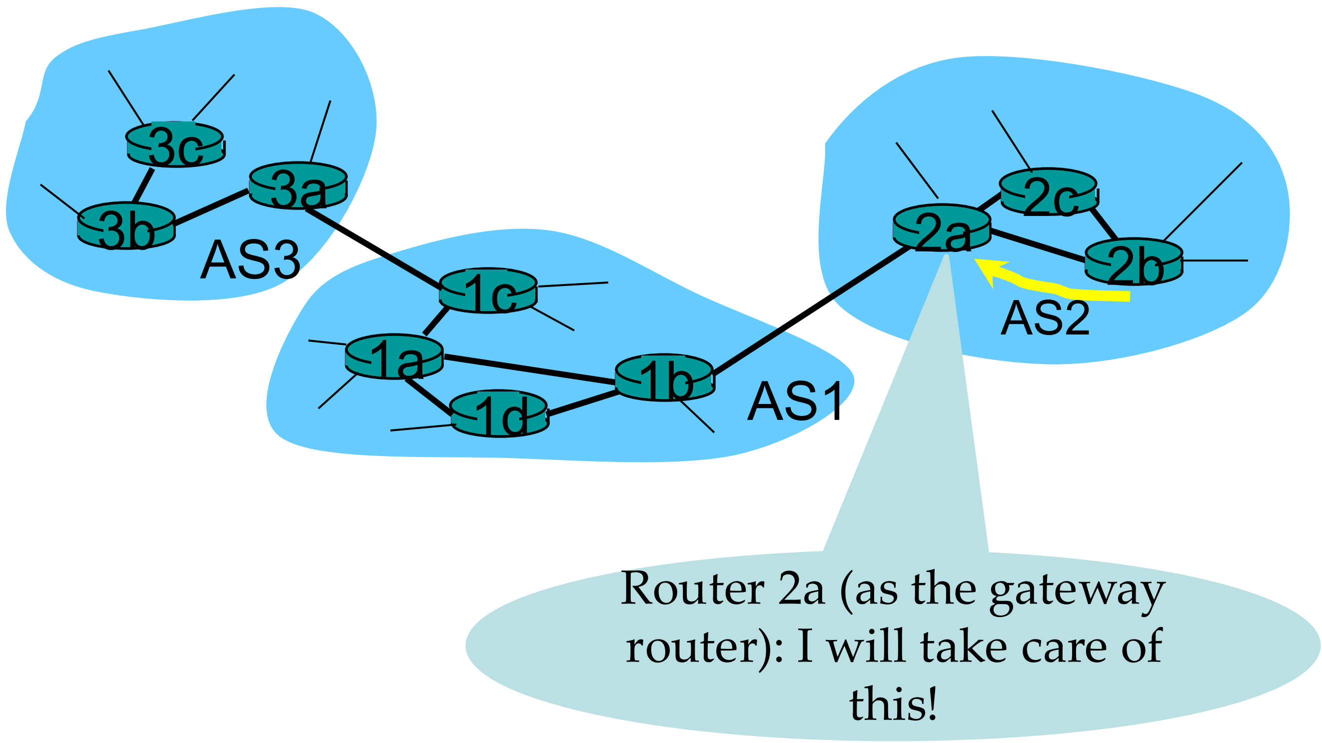
Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 3rd edition, 2005, Figure 4.17.

# Interconnected Autonomous Systems

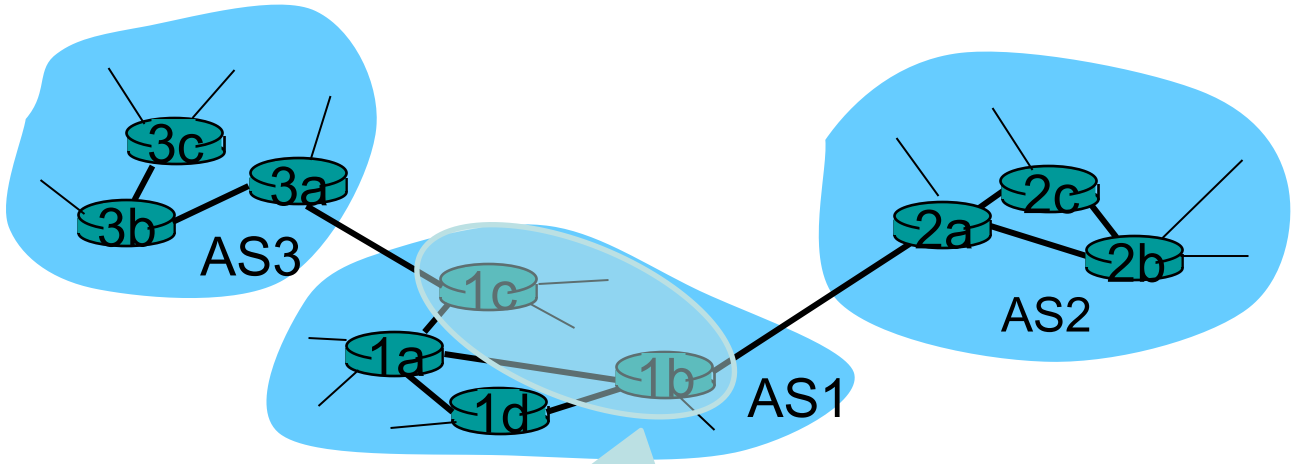



How does a router know how to route a packet to a destination that is outside the AS?

# Case 1: Only one gateway router to only one other AS



# Case 2: Two or more links to the other ASs

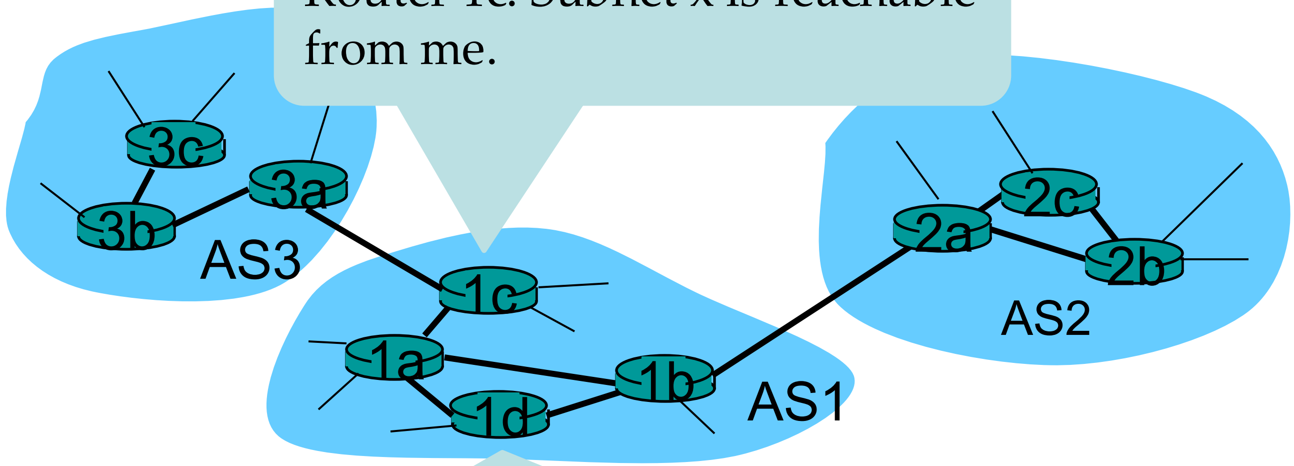



**Handled by the inter-AS routing protocol**
**Routers 1b and 1c:**

(1) obtain the reachability information from the neighbouring ASs
(2) propagate the reachability information to all routers inside the AS

# Case 2: Two or more links to the other ASs (cont.)

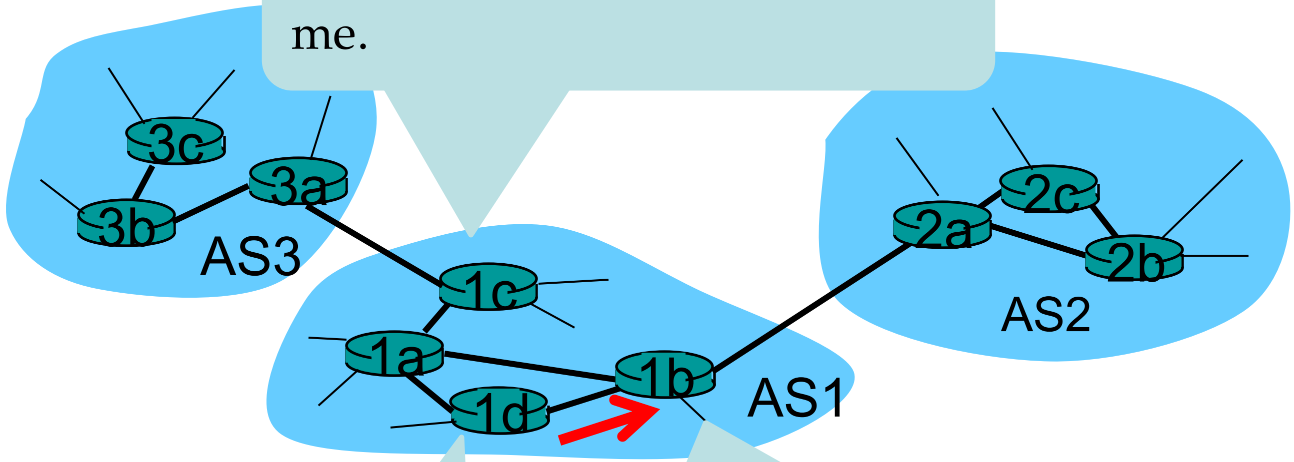
Router 1c: Subnet x is reachable from me.




Router 1d:
(1) Determine the router interface on the least-cost path to gateway router 1c from the intra-AS routing protocol
(2) Add an entry *(x, I)* into its forwarding table

# Case 2: Two or more links to the other ASs (cont.)

1c: Subnet x is reachable from me.




1b: Subnet x is reachable from me.

1d*: Which way to go*? **Hot-potato routing** – using the intra-routing to choose the nearest gateway router

# Main steps in adding an outside-AS destination in a router's forwarding table

Learn from **inter-AS routing protocol** that subnet x is reachable via multiple gateways.

Use routing info from **intra-AS routing protocol** to determine costs of **least-cost paths to each of the gateways**.

**Hot-potato routing**: Choose the least-cost gateway from intra-AS routing protocol.

Determine from **intra-AS routing protocol** the **router interface** I that leads to least-cost gateway. Enter (x,I) in forwarding table.

# Inter-Autonomous System Routing

# Border Gateway Protocol (BGP)

The *de facto* exterior router protocol for the Internet

Allow routers/gateways in different autonomous systems (ASs) to cooperate in the exchange of routing information

Use a modified distance vector algorithm, referred to as a "Path Vector" algorithm

The current version of BGP known as BGP-4 (RFC 4271)

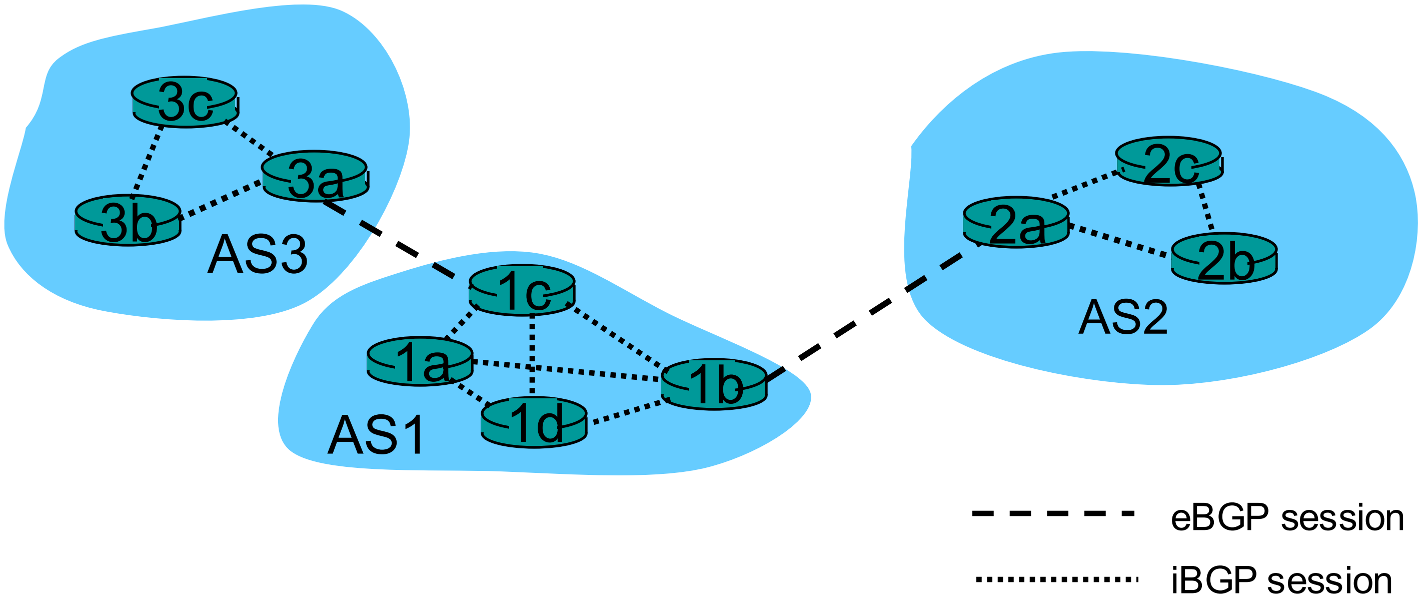
Source: https://datatracker.ietf.org/doc/html/rfc4274

# BGP Basics

- BGP provides each AS a means to:
  1. Obtain the subnet reachability information from neighbouring ASs
  2. Propagate the reachability information to all routers internal to the AS
  3. Determine “good” routes to subnets based on the reachability information and AS policy

# BGP Basics (cont.)

- Pairs of routers (BGP peers) exchange routing information over TCP connections.
- The TCP connection along with all the BGP messages sent over the connection is a BGP session.
- Destinations are CIDRised prefixes, with each prefix representing a subnet or a collection of subnets.

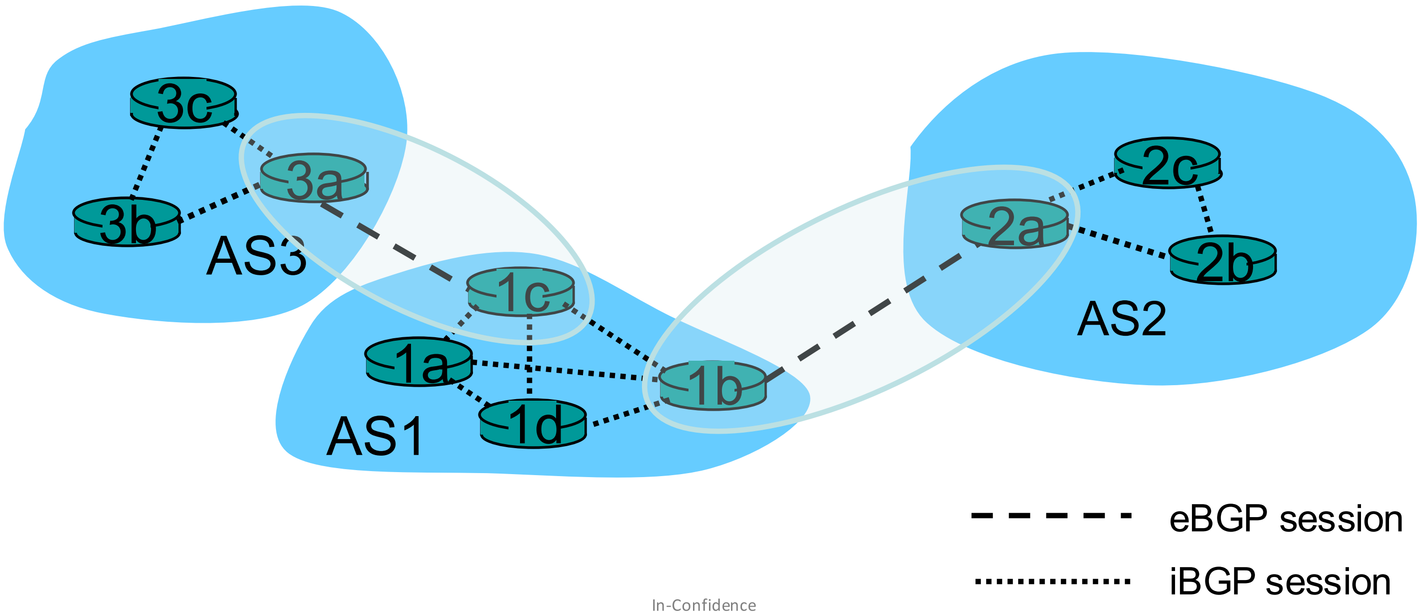


# Functional Procedures

- Neighbour acquisition
  - Two neighbouring routers in different autonomous systems agree to exchange routing information regularly.

**Open messages:** open a neighbour relationship with another router

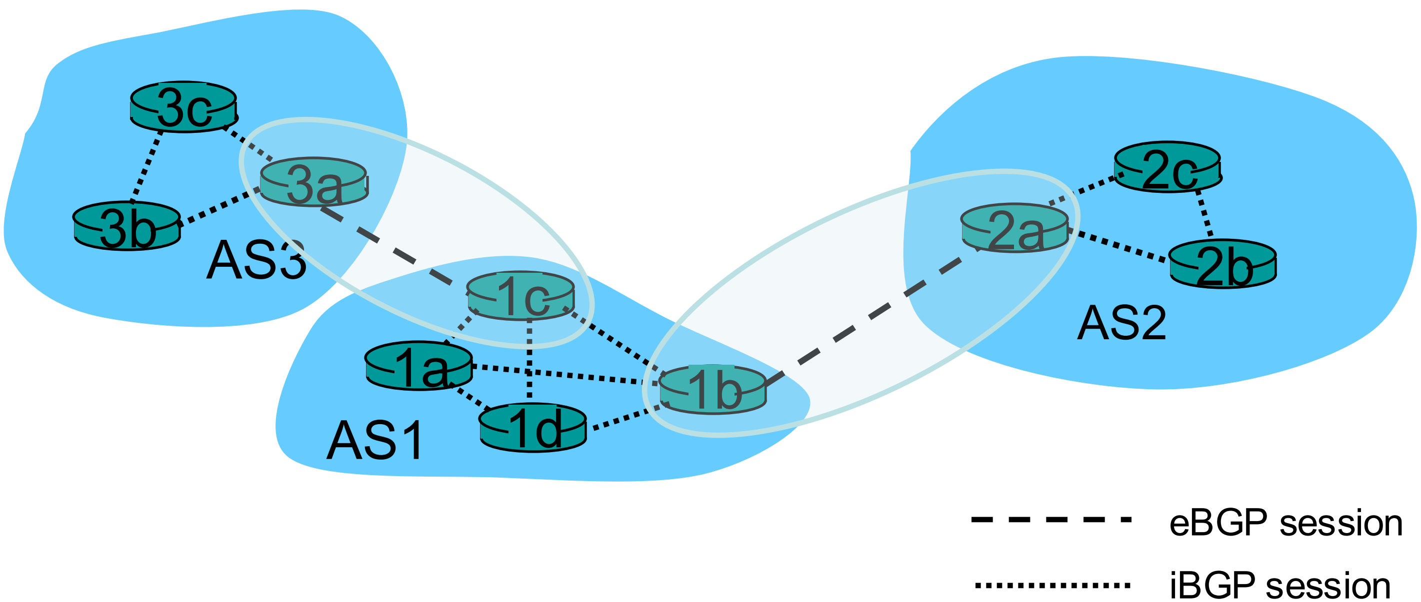
**Keepalive messages:** acknowledge an Open message



# Functional Procedures (cont.)

- Neighbour reachability
  - Each partner needs to be assured that the other partner still exists and is still engaged in the neighbour relationship.

**Keepalive messages:** periodically confirm the neighbour relationship

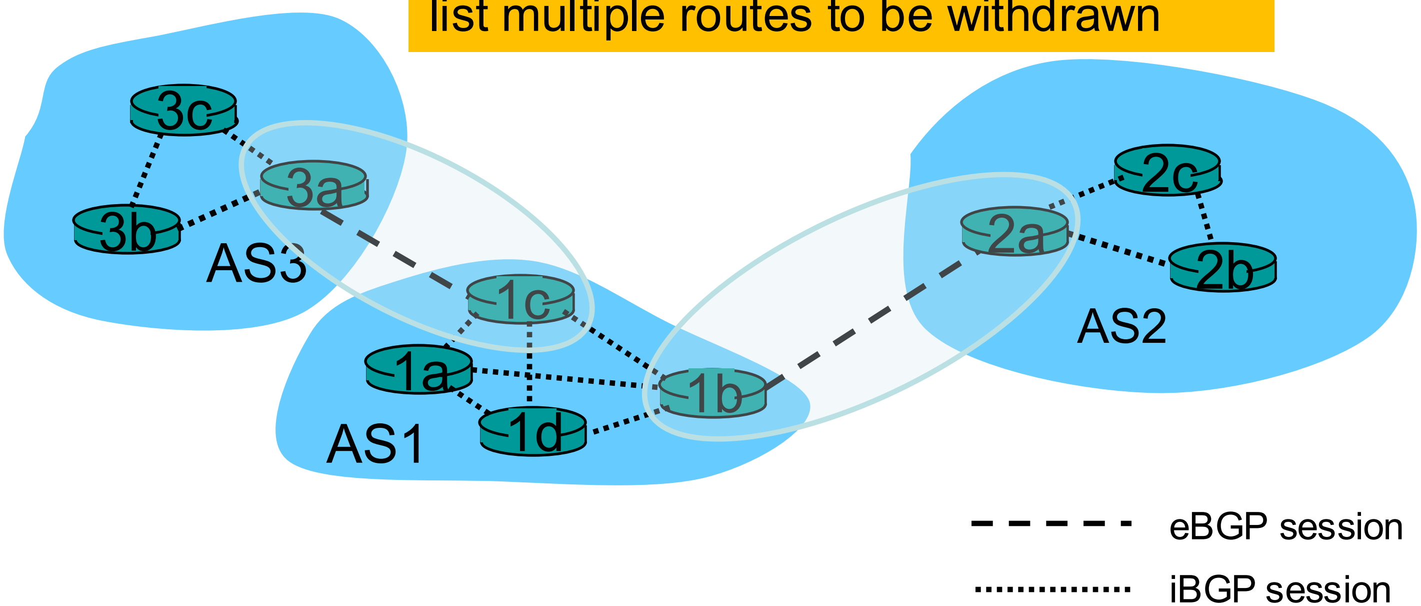



In-Confidence

# Functional Procedures (cont.)

- Network reachability
  - Each router maintains a database of the networks that it can reach and the preferred route for reaching each network.

**Update messages:** transmit information about a single route and/or list multiple routes to be withdrawn



# Path Attributes

- When a router advertises a prefix, it includes with the prefix a number of BGP attributes.

- Two important attributes:

  - AS_Path: contains the ASs through which the advertisement for the prefix passed; if a router sees its AS is contained in the path list, it rejects the advertisement (to avoid loop).

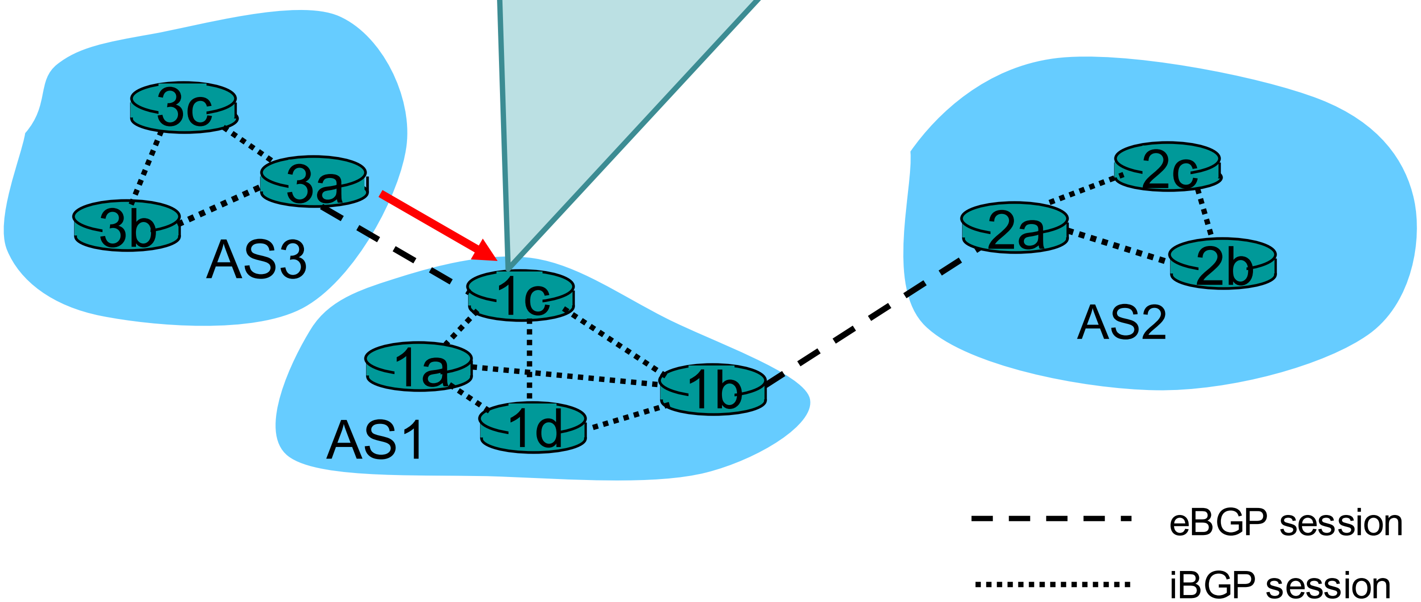
  - Next_Hop: indicates the specific gateway router of the AS (which sends the advertisement).

- When a gateway router receives a router advertisement, it uses its import policy to decide whether to accept or filter the route.

# BGP Routing Information Exchange - Example

- Router 3a in AS3 can issue an Update message to router 1c in AS1.

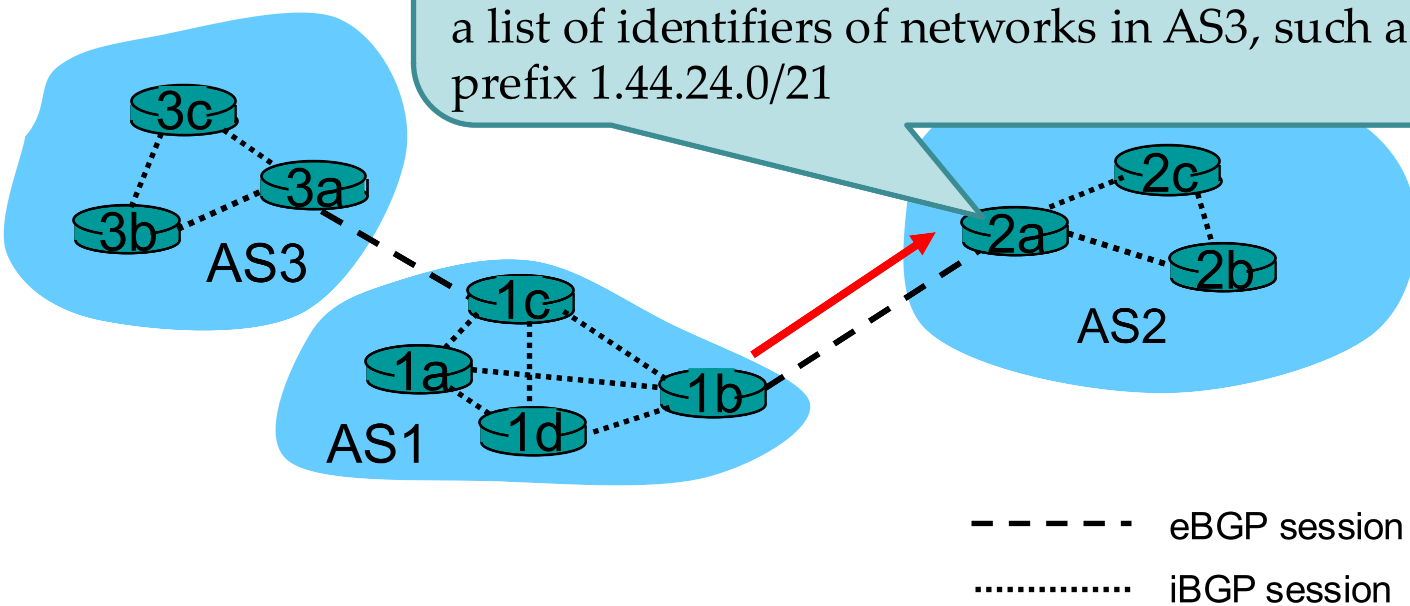
**UPDATE msg received by 1c:**
**AS_Path:** {AS3}
**Next_Hop:** IP address of 3a
**NLRI (Network Layer Reachability Information):** a list of identifiers of networks in AS3, such as a prefix 1.44.24.0/21



# BGP Routing Information Exchange – Example (cont.)

- Router 1c can use iBGP sessions to distribute this new reachability information to all routers in AS1.
- Router 1b can then advertise the new reachability information to AS2 (using eBGP sessions).
- When router 2a learns about a new prefix, it creates an entry for the prefix in its forwarding table.

**UPDATE MSG received by 2a:**
**AS_Path:** {AS1, AS3}
**Next_Hop:** IP address of 1b
**NLRI (Network Layer Reachability Information):** a list of identifiers of networks in AS3, such as a prefix 1.44.24.0/21



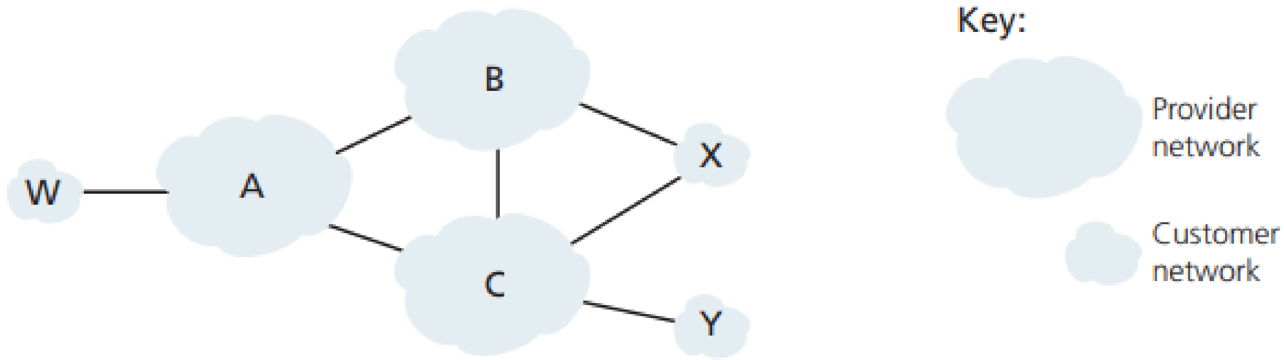
# BGP Route Selection

- Each BGP router may learn about more than one route to any one prefix and must select one of the possible routes.

- Elimination rules:

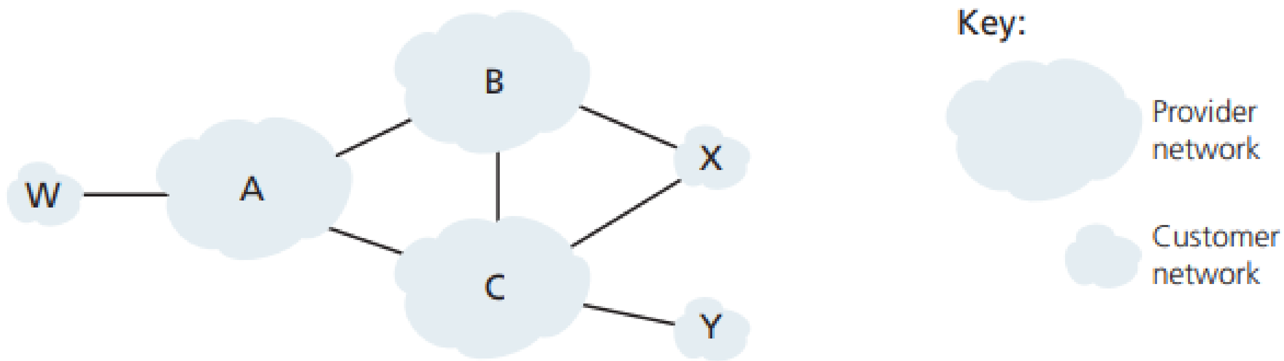
1. Highest local preference value (assigned as an attribute)
2. Shortest AS-PATH
3. Closest NEXT-HOP router (hot-potato routing)
4. Additional criteria

# BGP Routing Policy (cont.)




- A advertises to B a path AW; B advertises to X the path BAW
- Should B advertise to C the path BAW?
  - If yes, B gets no “revenue” for routing YCBAW since neither W nor Y is B’s customer. (free riding!)
  - B may force C to route to/from A’s customers via a direct connection between A and C.
  - *Rule of thumb (typically for tier-3 and many tier-2 ISPs)* – Any traffic flowing across an ISP’s backbone network must have either a source or a destination (or both) in a network that is a customer of that ISP.

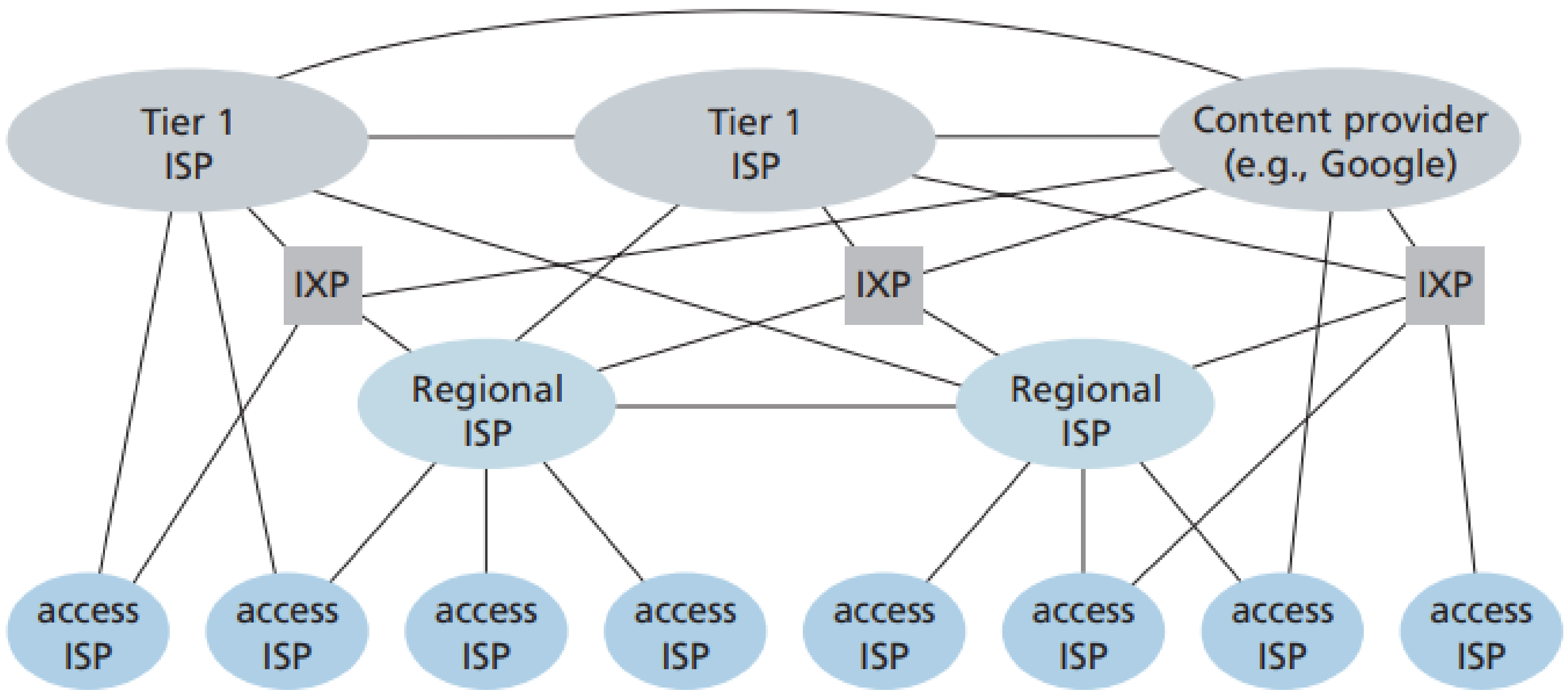
# BGP Routing Policy




- X is dual-homed (multi-homed)
- How will X be prevented from forwarding traffic between B and C?
  - X functions as a stub network and advertises that it has no paths to any other destinations except itself

Selective routing advertisement policy

# ISP Hierarchy (Optional Content)




Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, 8th edition (global edition), 2022, Figure 1.15.

# Intra-Autonomous System Routing

# Intra-AS Routing

- Also known as Interior Router Protocols
- Most common Intra-AS routing protocols
  - RIP: Routing Information Protocol
  - OSPF: Open Shortest Path First
  - IS-IS: Intermediate System to Intermediate System (OSPF's closely related cousin)
  - IGRP: Interior Gateway Routing Protocol (Cisco proprietary)

# Routing Information Protocol (RIP)

One of the oldest distance vector routing protocols

Based on the Bellman-Ford algorithm

Use the hop count as the distance metric

The current version of RIP known as RIP version 2 (RFC 2453)

Source: https://datatracker.ietf.org/doc/html/rfc2453

# RIP Basics

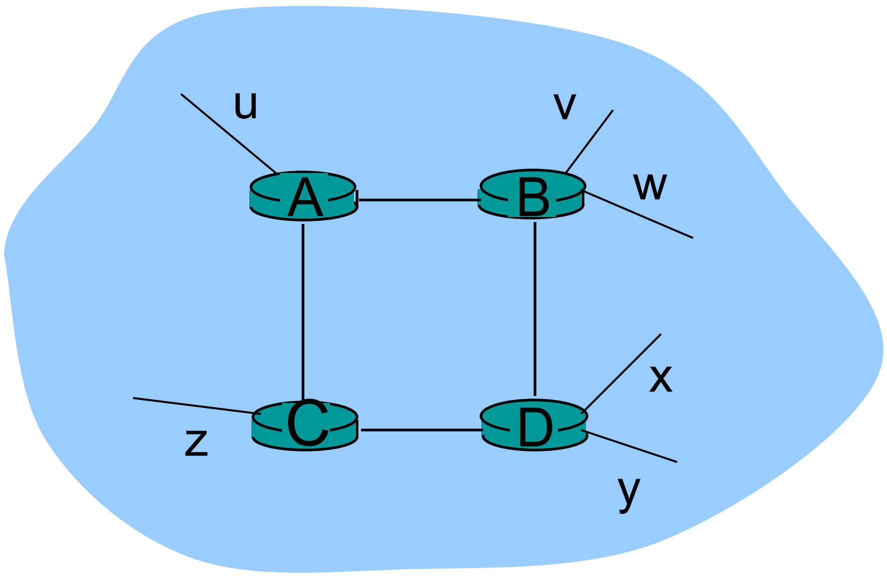
- A Request message is used to ask for a response containing all or part of a router’s RIP table.
  - RIP table, known as a routing table, includes both the router’s distance vector and its forwarding table

- A Response message (also known as a RIP advertisement) is for one of several different reasons:
  - Response to a specific query
  - Regular update (unsolicited response)
  - Triggered update caused by a route change

Source: https://datatracker.ietf.org/doc/html/rfc2453

# RIP Basics (cont.)

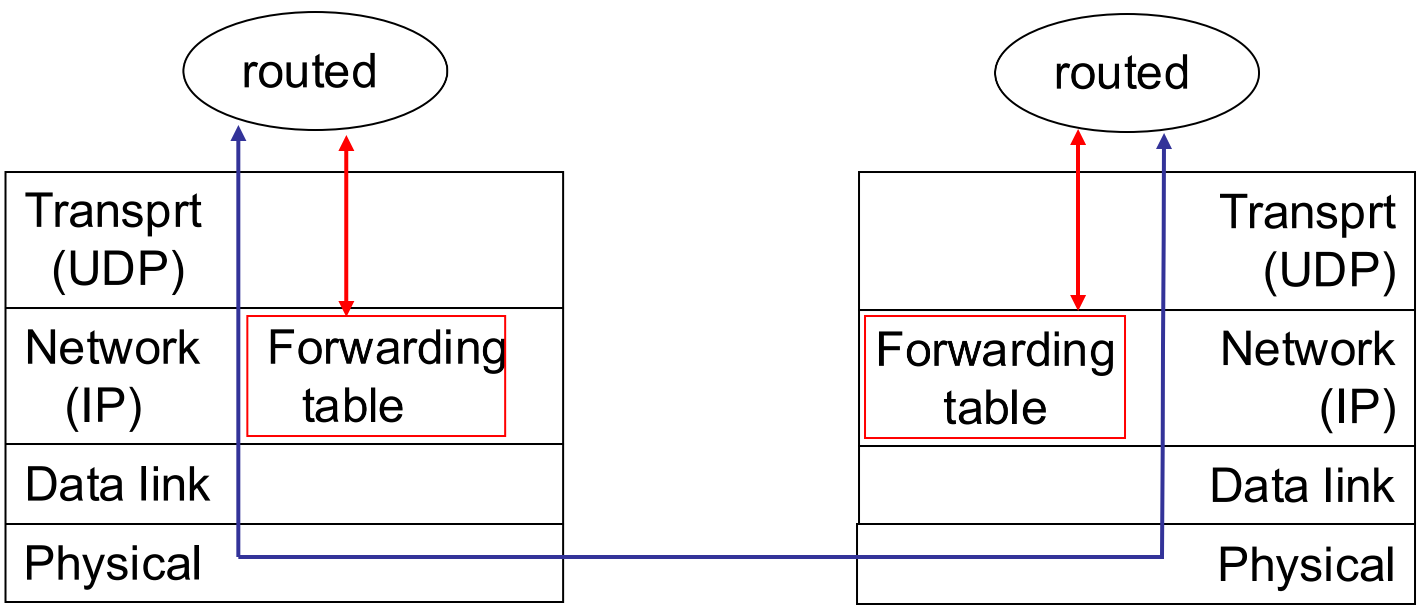
- Distance metric: # of hops (**max = 15 hops**)
  - # of hops: # of subnets traversed along the shortest path from the source router to the destination subnet, including the destination subnet (e.g., source = A)




| destination | hops |
|---|---|
| u | 1 |
| v | 2 |
| w | 2 |
| x | 3 |
| y | 3 |
| z | 2 |

# RIP Basics (cont.)

- RIP is a UDP-based protocol (port number 520)
- RIP is typically implemented as an application-layer process



# RIP Basics (cont.)

- Each router sends its advertisement every 30 seconds (or whenever its RIP table changes) to all its neighbours via a Response Message.
  - Each response message contains a list of up to 25 destination subnets within the AS and the sender's distance to each of those subnets.
- The route is considered to have expired upon the timeout (180 seconds).
- The route is removed from the RIP table upon expiration of the garbage-collection timer (120 seconds).

# RIP Solutions to the Count-to-Infinity Problem

Count-to-16

- Maximum number of hops bounded to 15
- A metric of infinity is set to 16

Split horizon

- Don't send a neighbour the routing information learned from this neighbour

Split horizon with poison reverse

- Send the routing information learned from this neighbour but set their metrics to infinity

# Open Shortest Path First (OSPF)

Open: published in the open literature

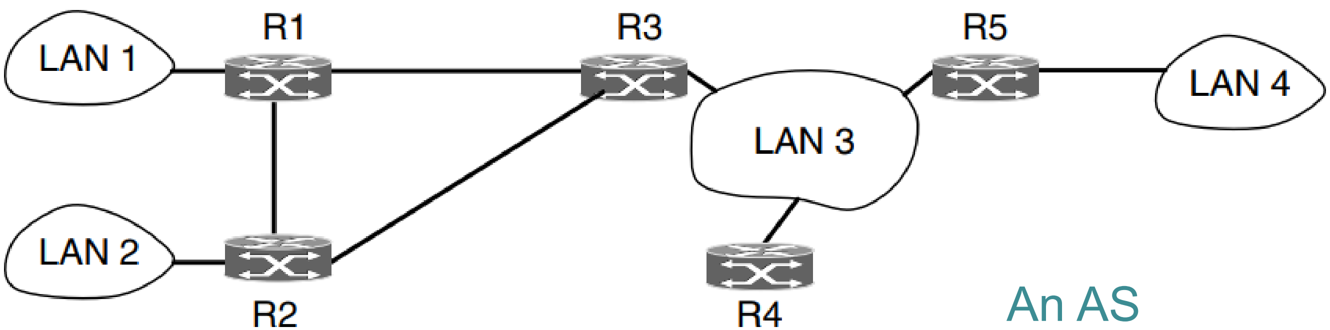
Use the link-state routing algorithm

Advertisements disseminated to the entire AS (via broadcasting) and carried directly by IP

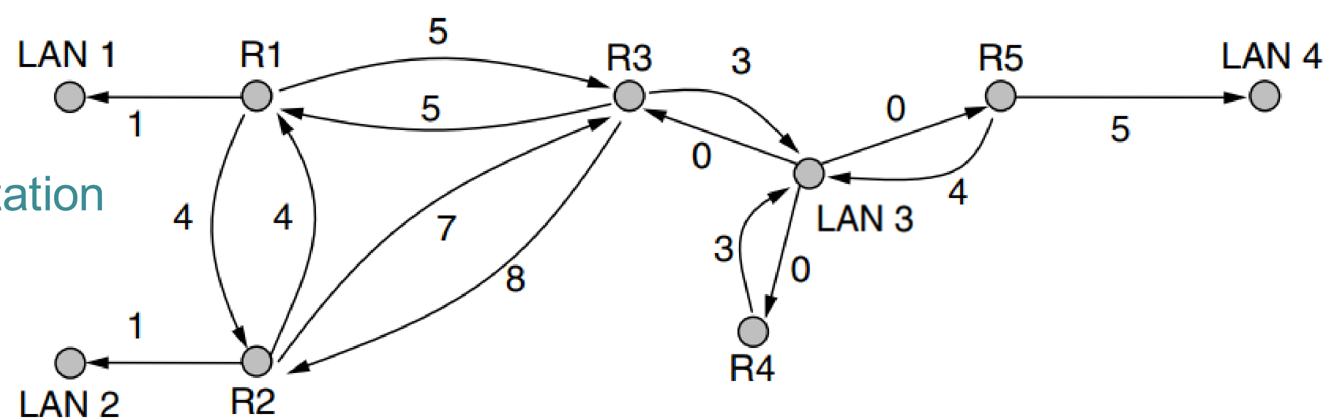
The current version of OSPF known as OSPF version 2 (RFC 2328)

# OSFP Basics

- Each router maintains a database that reflects the known topology of the autonomous system of which it is a part.




An AS




A graph representation of the AS

Source: Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, 5th edition, 2010, Figure 5-64.

# OSPF Basics (cont.)

## Security

- All OSPF protocol exchanges are authenticated.

## Equal-cost multipath (ECMP)

- OSPF remembers the set of shortest paths and splits the traffic across them during packet forwarding to balance load.

## Traffic engineering

- OSPF computes a route through the internet that incurs the least cost based on a user-configurable metric of cost.

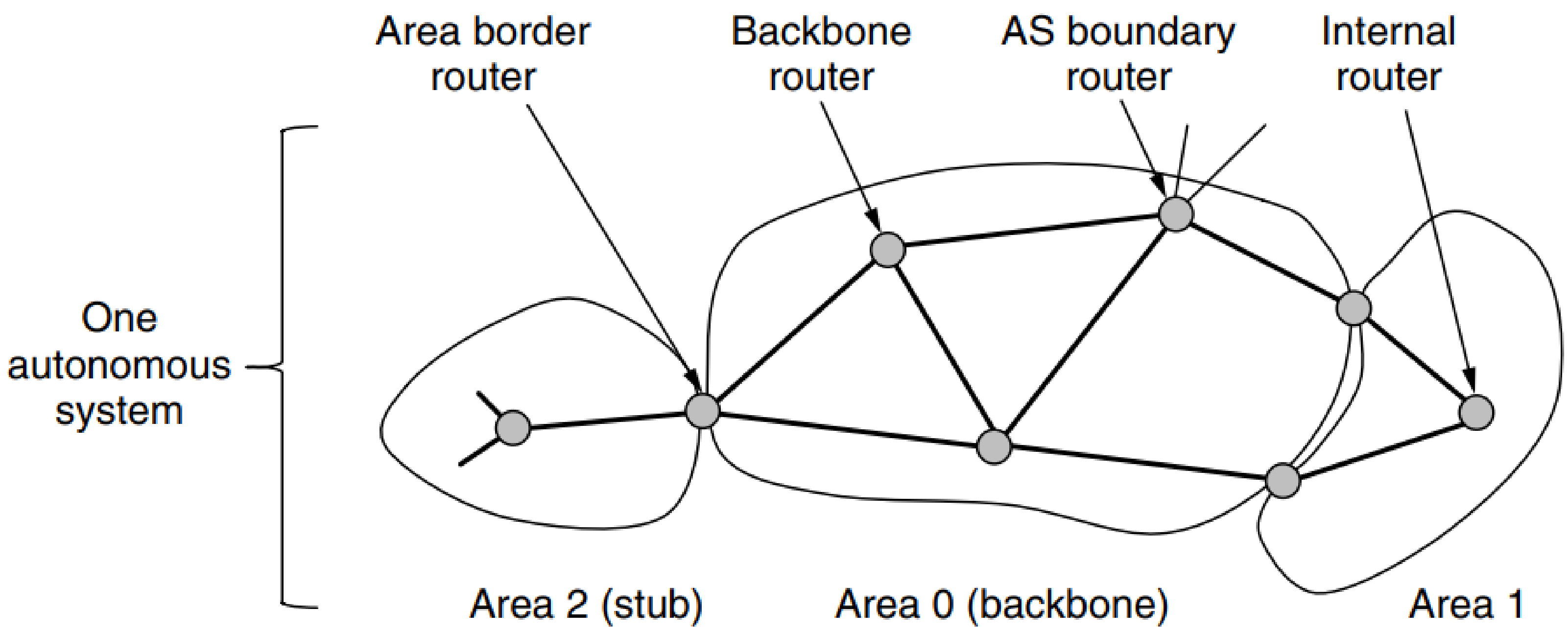
## Area routing

- OSPF allows an AS to be divided into numbered areas.

Source: https://info.support.huawei.com/info-finder/encyclopedia/en/OSPF.html

# Hierarchical OSPF

- An area is a network or a set of contiguous networks.
- Each area runs a separate copy of the OSPF algorithm.




Source: Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, 5th edition, 2010, Figure 5-65.

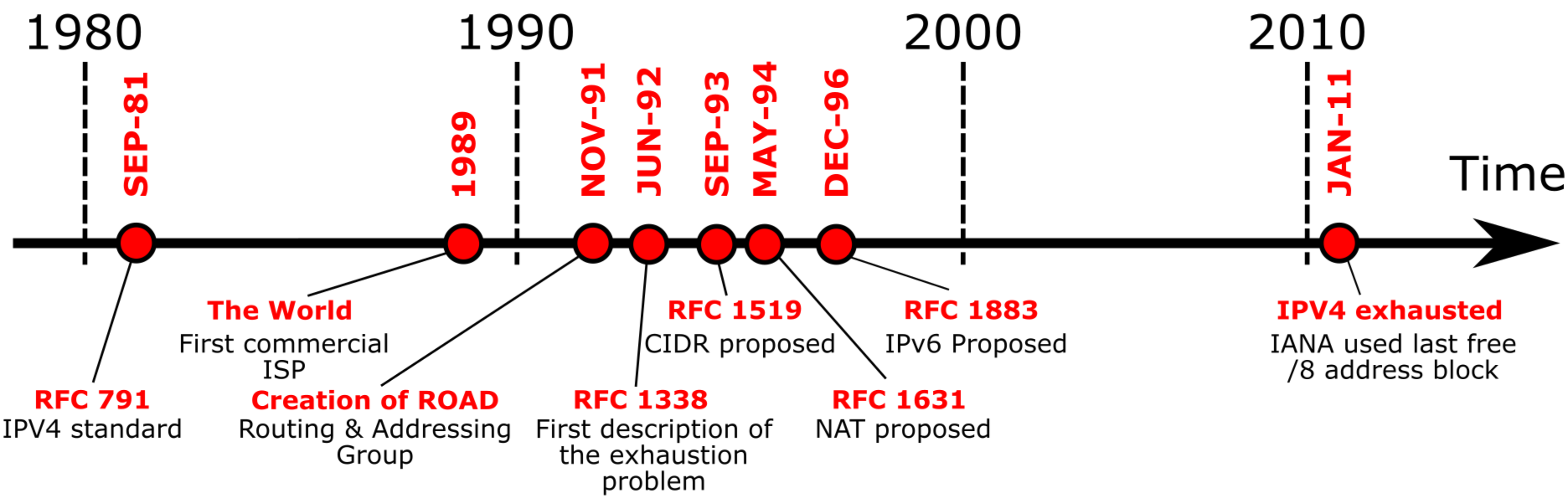
# Hierarchical OSPF (cont.)

- There can be multiple AS boundary routers in an AS.
- AS boundary routers may be internal or area border routers and may or may not participate in the backbone.

# IP Addresses

# IP addresses are scarce

- IPv4 address exhaustion timeline




Source: https://en.wikipedia.org/wiki/IPv4_address_exhaustion

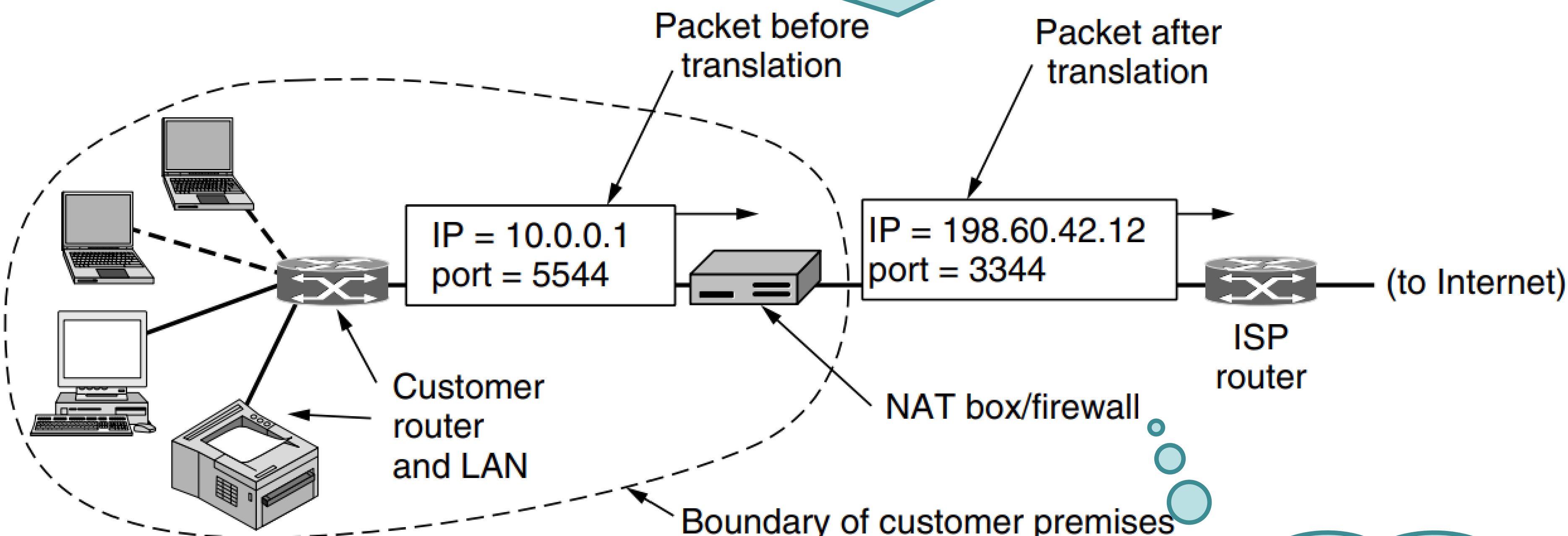
# Network Address Translation (NAT)

- The ISP assigns each home or business a single IP address (or at most, a small number of them) for Internet traffic.


Source: https://www.cisco.com/c/en/us/products/routers/network-address-translation.html

# The Operation of NAT

Before a packet exits the local network and goes to the ISP, an address translation from the unique internal IP address to the shared public IP address takes place.




Every host gets a unique IP address through DHCP (Dynamic Host Configuration Protocol).

How does the NAT box know which internal address to replace it with?

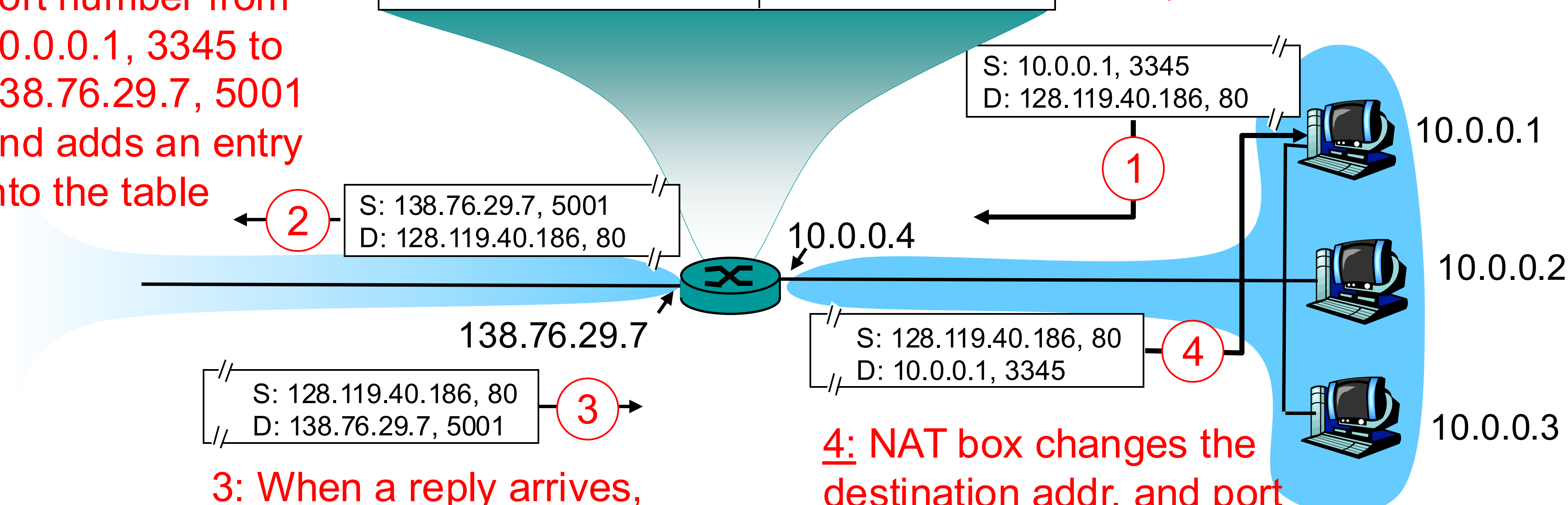
Source: Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, 5th edition, 2010, Figure 5-55.

# The Operation of NAT (cont.)

| NAT translation table | |
|---|---|
| WAN side addr. | LAN side addr. |
| 138.76.29.7, 5001 | 10.0.0.1, 3345 |
| …… | …… |

1: Host 10.0.0.1 sends a packet to a web server (128.119.40.186, 80)

2: NAT box changes the source addr. and port number from 10.0.0.1, 3345 to 138.76.29.7, 5001 and adds an entry into the table




3: When a reply arrives, NAT box indexes the NAT translation table using the destination addr. and port number

4: NAT box changes the destination addr. and port number from 138.76.29.7, 5001 to 10.0.0.1, 3345

# Motivations of NAT

- The local network uses just one IP address as far as the outside world is concerned
  - no need to be allocated a range of addresses from the ISP
  - can change addresses of devices in the local network without notifying the outside world
  - can change the ISP without changing addresses of devices in the local network
  - devices inside the local network not explicitly addressable, visible by the outside world

# An Example

| University of Canterbury | |
|---|---|
| **AS Handle** | AS9432 |
| **ASN Name** | CANTERBURY-AS |
| **Organization Name** | University of Canterbury |
| **Organization ID** | ORG-VNZL1-AP-APNIC |
| **Country** | New Zealand |
| **Regional Registry** | APNIC |
| **IPv4 CIDRs** | • 132.181.0.0/16 • 202.36.178.0/23 |

```
Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . : canterbury.ac.nz
   Link-local IPv6 Address . . . . . : fe80::84d1:5f80:5c58:7b6c%4
   IPv4 Address. . . . . . . . . . . : 10.34.27.79
   Subnet Mask . . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . . : 10.34.254.254
```

# NAT is controversial

NAT violates the architectural model of IP.

NAT breaks the end-to-end connectivity model of the Internet.

Some processes on the Internet are not required to use TCP or UDP.

NAT violates the most fundamental rule of protocol layering.

The port-number is a 16-bit field.

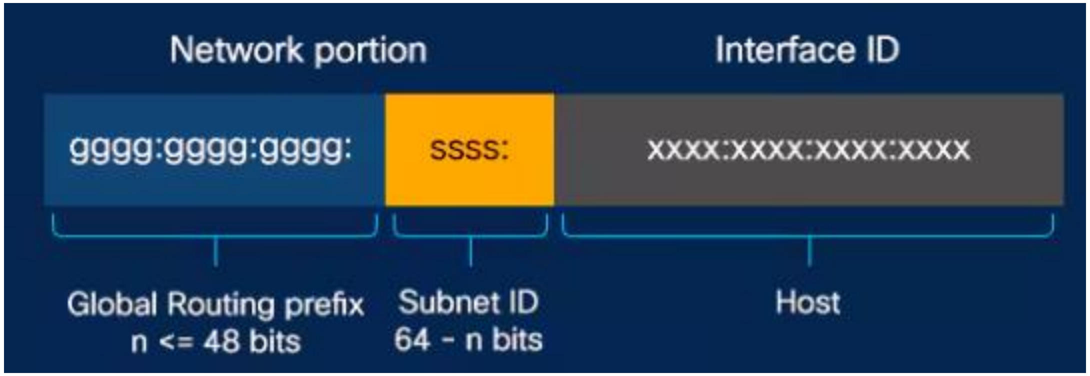
# CGNAT (Optional Content)

- Carrier-grade NAT, known as large-scale NAT

- Used by ISPs to manage the shortage of public IPv4 addresses
  - An ISP assigns private IP addresses (in the 100.64.0.0/10 range, as per RFC 6598) to multiple customers.
  - All customers behind the CGNAT share a smaller pool of public IP addresses.

# IPv6 at a glance

- Longer addresses (128 bits long)
- Simplification of the header (8 fields) for faster packet processing
- Better support for options through extension headers
- Improved QoS

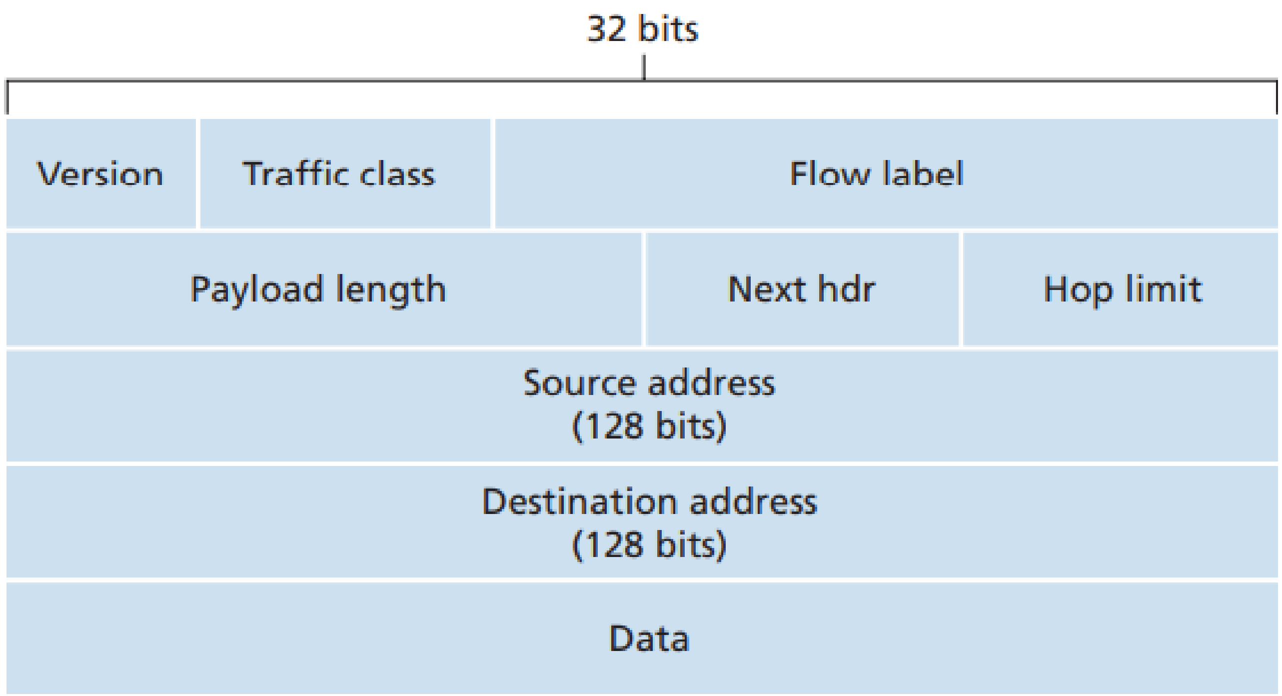



Source: https://www.cisco.com/c/en/us/solutions/ipv6/overview.html#~addressing-structure

# IPv6 Header Format

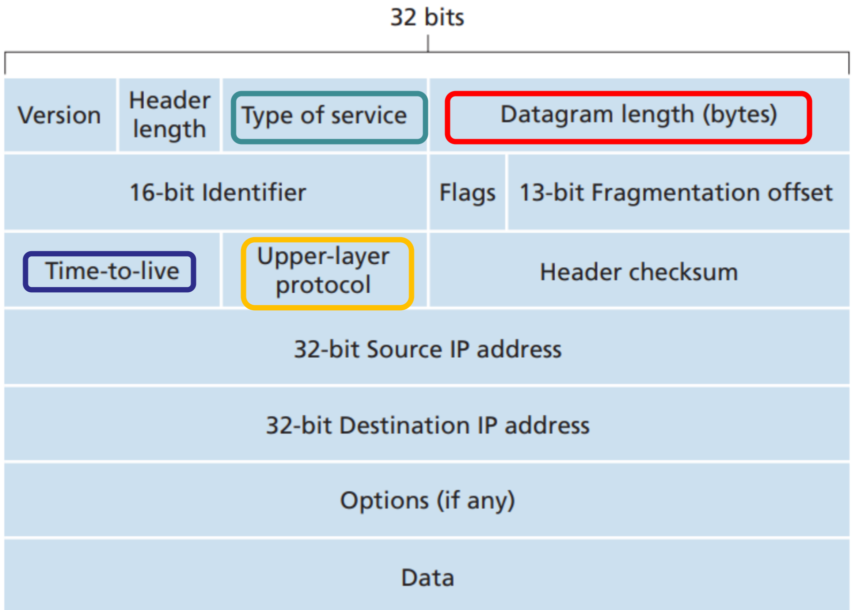
Flow Label: mark groups of packets

Next header: identify the type of header immediately following the IPv6 header




Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, Addison Wesley, 3rd edition, 2005, Figure 4.22

# IPv4 vs. IPv6

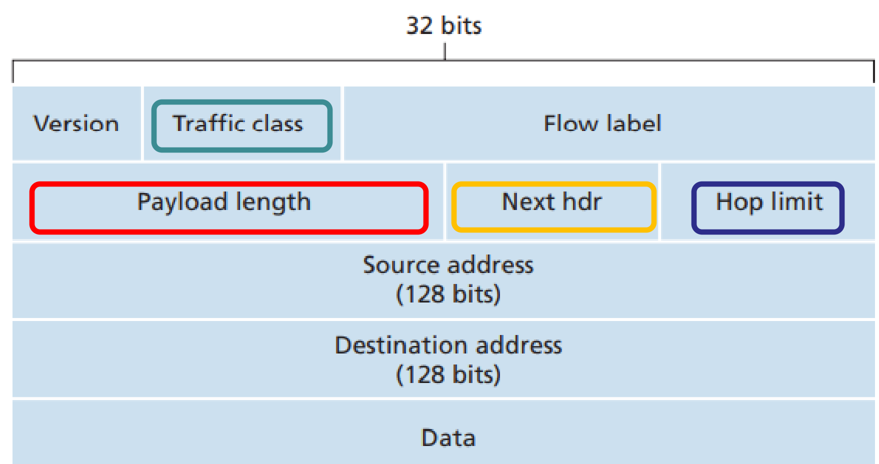



Facilitate QoS

IPv4: header + data
IPv6: data only

A field that is decremented on each hop

IPv4: the upper-layer protocol
IPv6: the upper-layer protocol or extension header




Source: James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, Addison Wesley, 3rd edition, 2005, Figure 4.13 & 4.22.

# Some Examples

```
+---------------+------------------------
|  IPv6 header  | TCP header + data
|               |
| Next Header = |
|      TCP      |
+---------------+------------------------


+---------------+----------------+------------------------
|  IPv6 header  | Routing header | TCP header + data
|               |                |
| Next Header = |  Next Header = |
|    Routing    |      TCP       |
+---------------+----------------+------------------------


+---------------+----------------+-----------------+-----------------
|  IPv6 header  | Routing header | Fragment header | fragment of TCP
|               |                |                 |  header + data
| Next Header = |  Next Header = |  Next Header =  |
|    Routing    |    Fragment    |       TCP       |
+---------------+----------------+-----------------+-----------------
```

Source: https://datatracker.ietf.org/doc/html/rfc8200#section-4

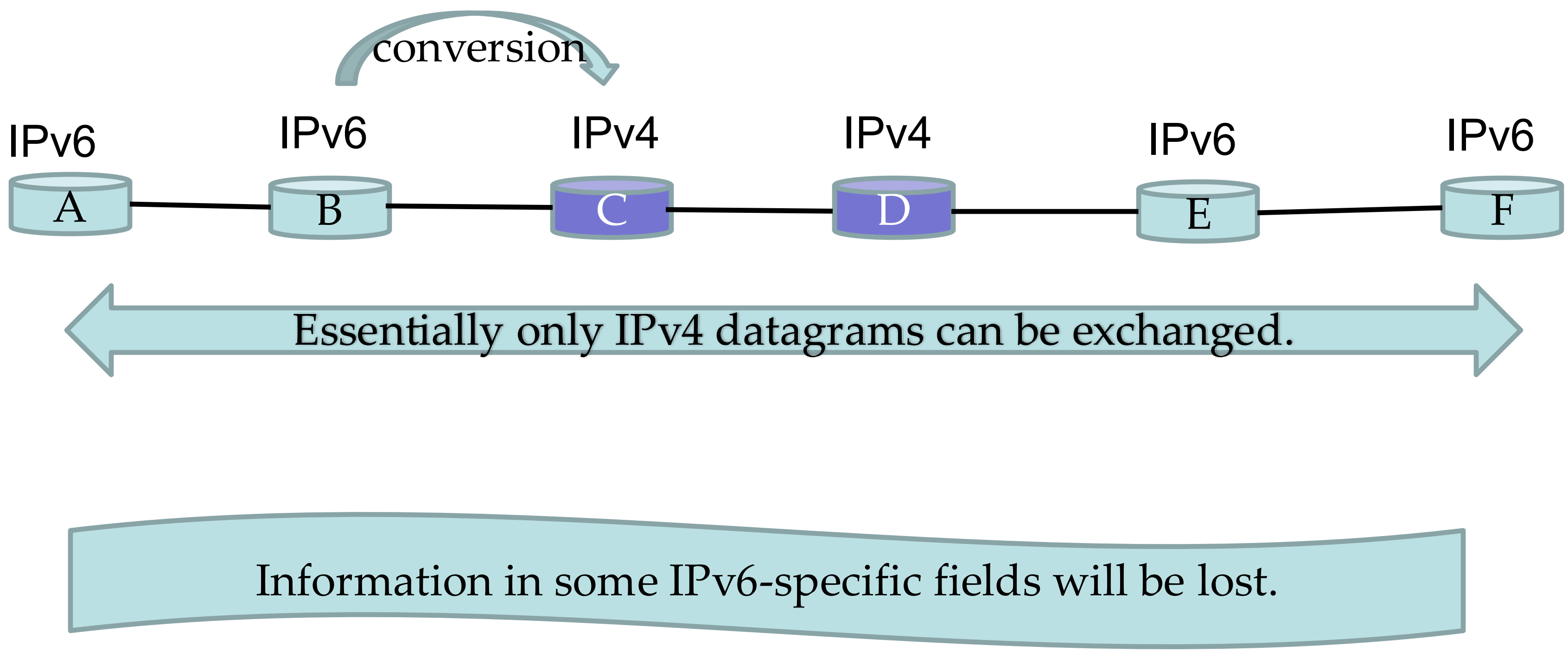
# Transition From IPv4 To IPv6

- How will the network operate with mixed IPv4 and IPv6 routers?
  - New devices are IPv4 and IPv6 capable; but old ones only supports IPv4.
  - Not all routers can be upgraded.

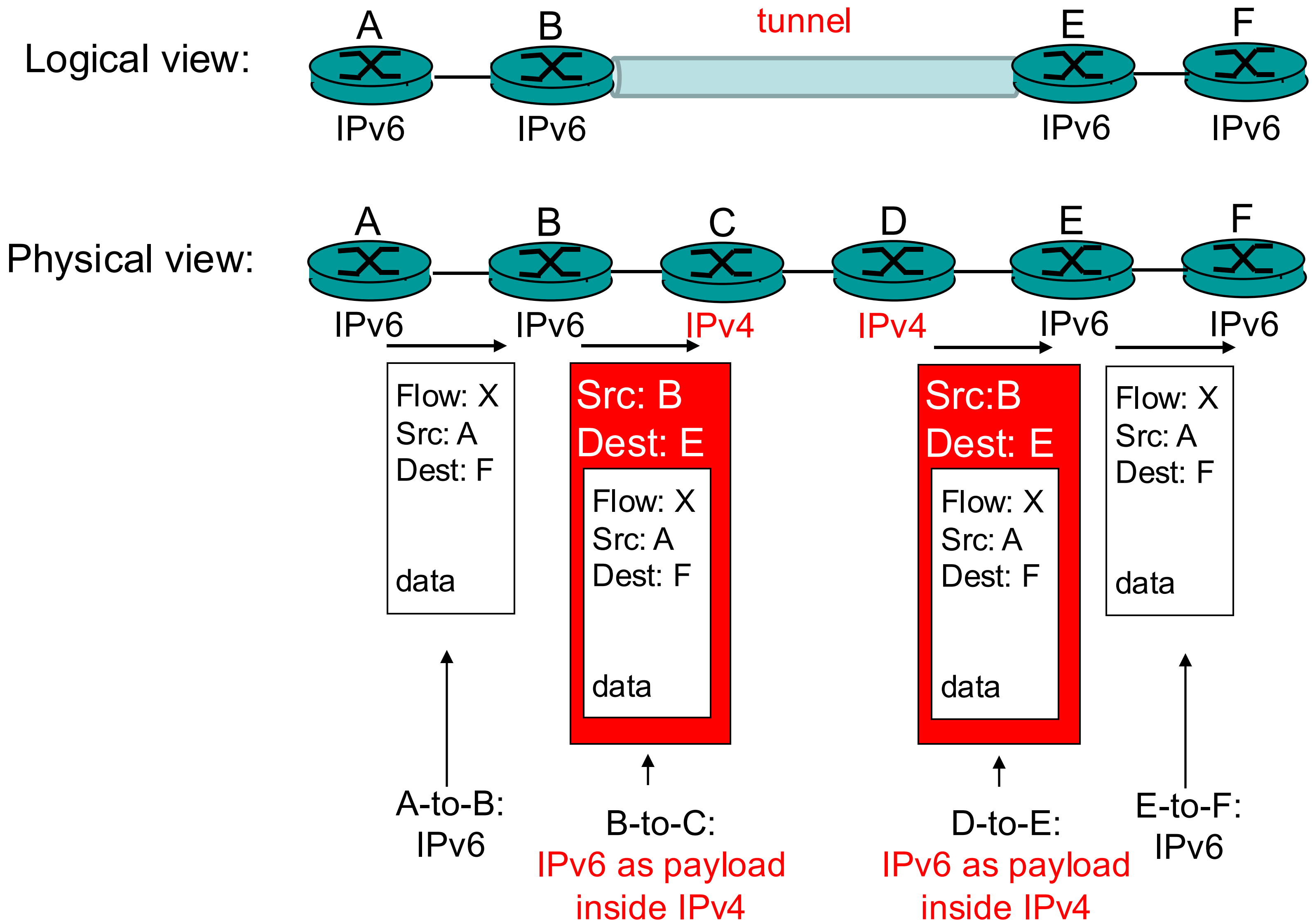
Source: https://datatracker.ietf.org/doc/html/rfc2893

# Dual-Stack Approach

- All IPv6 nodes also have a complete IPv4 implementation.

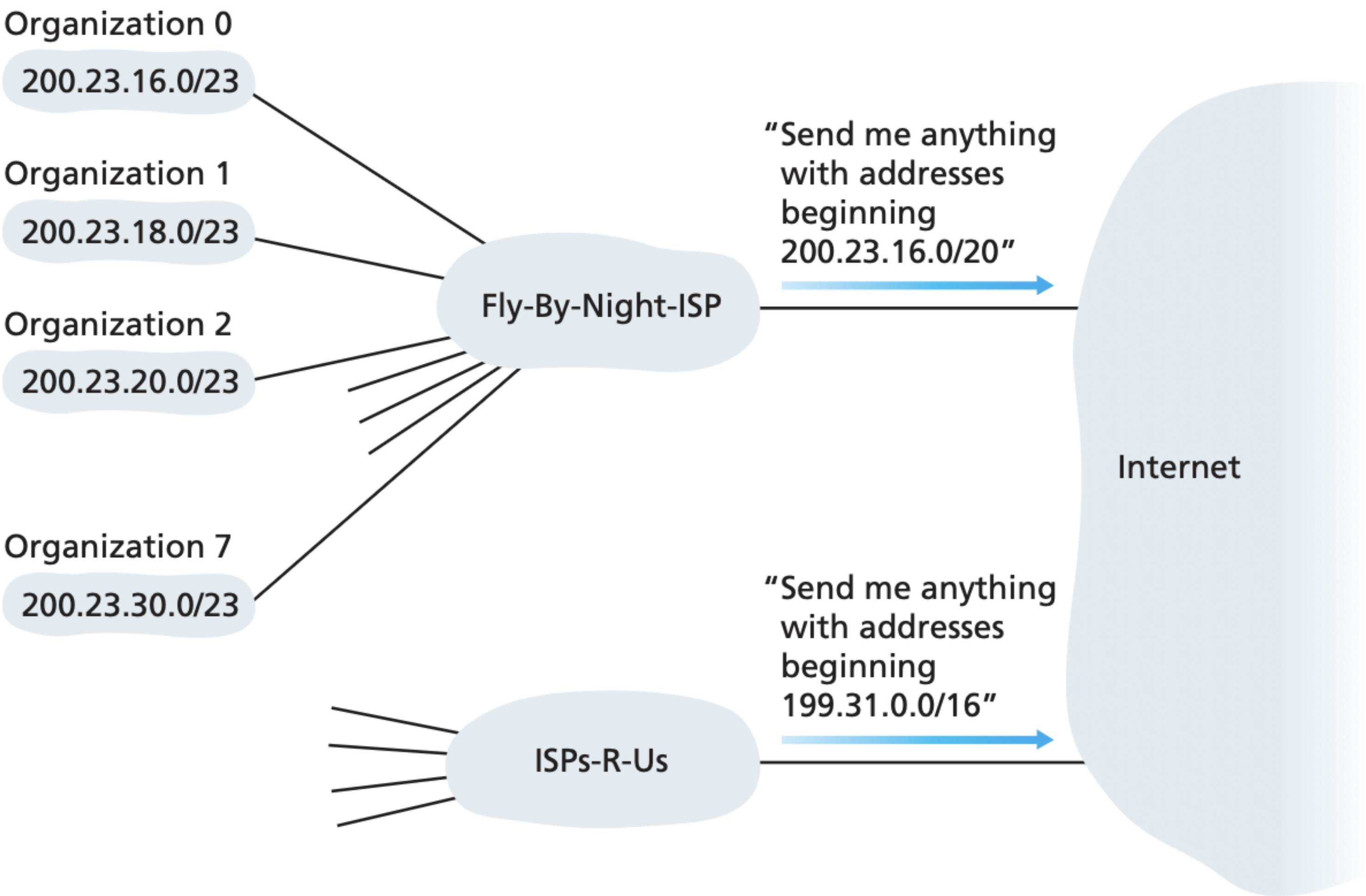


# Tunnelling



Logical view:
A
B
tunnel
E
F
IPv6
IPv6
IPv6
IPv6
Physical view:
A
B
C
D
E
F
IPv6
IPv6
IPv4
IPv4
IPv6
IPv6
Flow: X
Src: A
Dest: F
data
Src: B
Dest: E
Flow: X
Src: A
Dest: F
data
Src:B
Dest: E
Flow: X
Src: A
Dest: F
data
Flow: X
Src: A
Dest: F
data
A-to-B:
IPv6
B-to-C:
IPv6 as payload inside IPv4
D-to-E:
IPv6 as payload inside IPv4
E-to-F:
IPv6

# Address aggregation



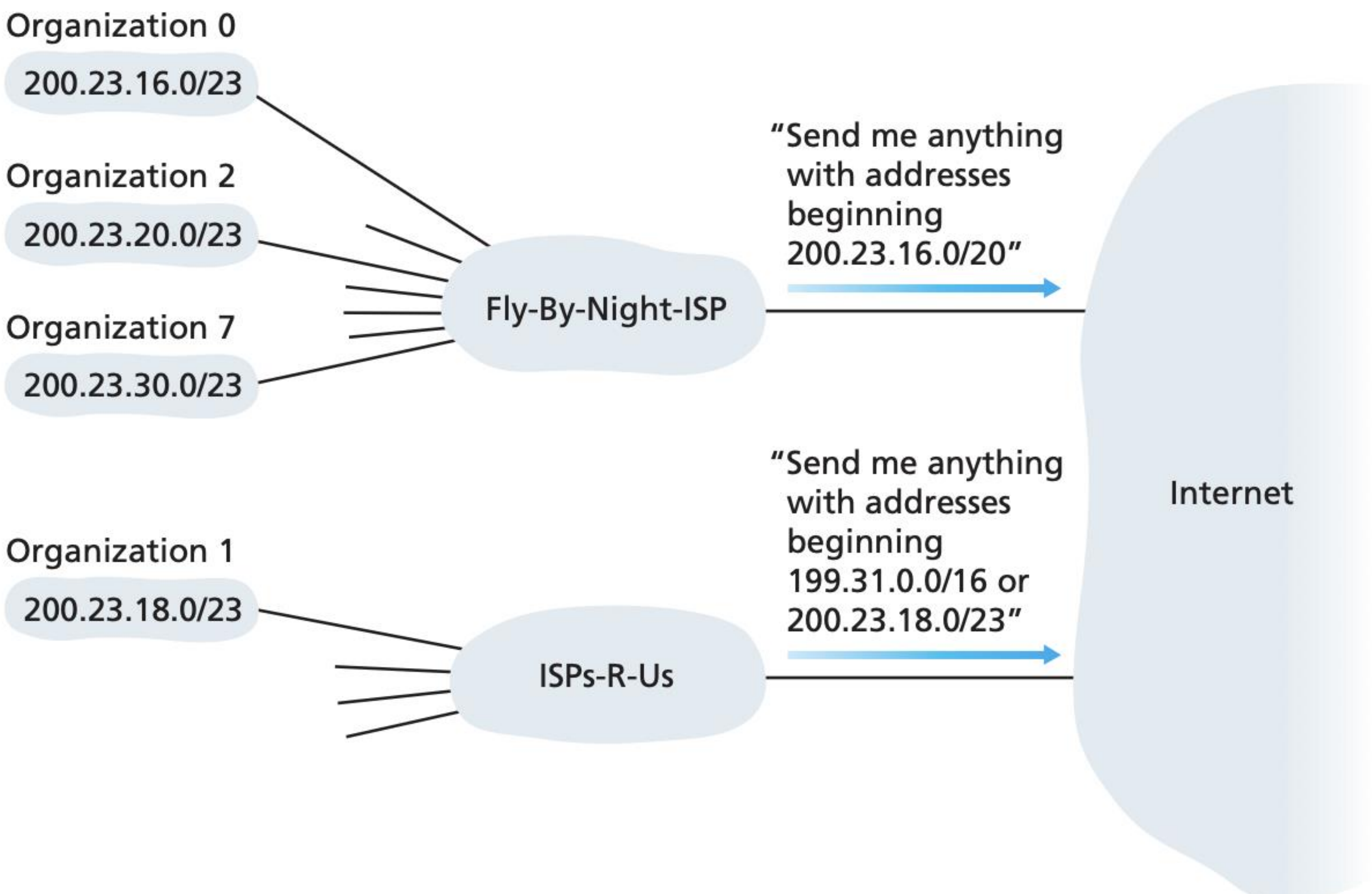

Source: [Kurose & Ross]

# Address aggregation

| | | |
|---|---|---|
| ISP's block | 200.23.16.0/20 | 11001000 00010111 00010000 00000000 |
| Organization 0 | 200.23.16.0/23 | 11001000 00010111 00010000 00000000 |
| Organization 1 | 200.23.18.0/23 | 11001000 00010111 00010010 00000000 |
| Organization 2 | 200.23.20.0/23 | 11001000 00010111 00010100 00000000 |
| . . . | . . . | . . . |
| Organization 7 | 200.23.30.0/23 | 11001000 00010111 00011110 00000000 |

Source: [Kurose & Ross]

# Address aggregation




Apply longest address prefix match if there are multiple matches in the forwarding table.

Source: [Kurose & Ross]

# Obtaining addresses

- An organization can obtain a block of IP addresses from its ISP;
- ISP obtains addresses from Internet registries (local: ARIN, RIPE, APNIC, LACNIC; or global: ICANN)
- A host typically obtains its IP address from a DHCP server or by manual configuration.

# References


- William Stallings, Data and Computer Communications, Pearson, 10th edition, 2013.
  - Chapter 19 Routing (19.3)
- Andrew S. Tanenbaum & David J. Wetherall, Computer Networks, Prentice Hall, 5th edition, 2010.
  - Chapter 5 The Network Layer (5.6)
- James F. Kurose & Keith W. Ross, Computer networking: a top-down approach featuring the Internet, Addison Wesley, 3rd edition, 2005.
  - Chapter 4 The Network Layer (4.6)

# Acknowledgements

- Lecture notes are developed based on slides from the following three sources:
  - Dr Mengmeng Ge’s lecture notes for COSC264, University of Canterbury, 2023
  - Assoc Prof Dan Kim’s lecture notes for COSC264, University of Canterbury, 2017
  - Prof Aleksandar Kuzmanovic’s lecture notes for CS340, Northwestern University, 2005, https://users.cs.northwestern.edu/~akuzma/classes/CS340-w05/lecture_notes.htm